STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: FAB-05 | TAG: BLAST-PRESS-10T20T

Abrasive Blast Cabinet & Hydraulic Shop Presses (Enerpac 10T & Redline 20T)

Facility Zone: Zone 5: Fabrication, Sheet Metal & Abrasive Finishing (FAB) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. Blast Cabinet: ANSI Z87.1 Safety Glasses + Integrated Cabinet Rubber Gloves
2. Hydraulic Presses: ANSI Z87.1 Goggles + Full Face Shield (Mandatory-Ballistic Shatter Hazard) + Steel-Toe Boots

2. PRIMARY OPERATIONAL HAZARDS

1. Hydraulic Press: Ballistic part shattering under 10-20 tons of load; high-pressure hydraulic fluid injection hazard from pinhole leaks
2. Blast Cabinet: Dust inhalation if opened before 30s dwell time; severe slip hazard from spilled glass bead media on shop floor

3. BEFORE OPERATION CHECKLIST

1. Blast Cabinet: Inspect integrated rubber gloves for holes; turn ON dust collector before blasting; verify air regulator <= 100 PSI.
2. Enerpac 10T & Redline 20T Presses: Verify bed support pins are fully seated through both sides of H-frame; inspect hoses, GA2 adapter, and GF10P pressure gauge; use ONLY rated V-blocks/mandrels (NEVER use hardened chrome tool sockets).

4. DURING OPERATION SAFETY RULES

1. Blast Cabinet: Blast at 45-60 deg angle away from viewing window and gloves.
2. Hydraulic Presses: Strictly obey the 'Center Rule' (align load vertically through center of gravity); stand to the side out of the 'Line of Fire' behind Lexan/mesh shield; STOP IMMEDIATELY if gauge pressure rises without ram movement.

5. AFTER USE & SHOWROOM CLEANUP

1. Blast Cabinet: WAIT 30 SECONDS DWELL TIME after blasting before opening cabinet doors; sweep any spilled floor media immediately.
2. Hydraulic Presses: Slowly open release valve to retract ram; never inspect hydraulic pinhole leaks with bare hands.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Presses: Stop pumping immediately, step out of Line of Fire, and slowly crack release valve.
Blast Cabinet: Release foot pedal and shut off orange air valve.